

Rail Fence

1. Encrypt the message "Enjoy your Journey" using Rail Fence with depth 2 and 3.

Depth 2

E	J	Y	O	R	O	R	E	Y
	N	O	Y	U	J	U	N	

Cipher: E J Y O R O R E N O Y U J U N Y

Depth 3

Cipher: E Y R R N O Y U J U N Y J O O E

2. Encrypt the message as per your choice using Rail Fence with depth 2 & 3

Depth 2

G	O	D	Y
	O	D	A

plaintext: Good day

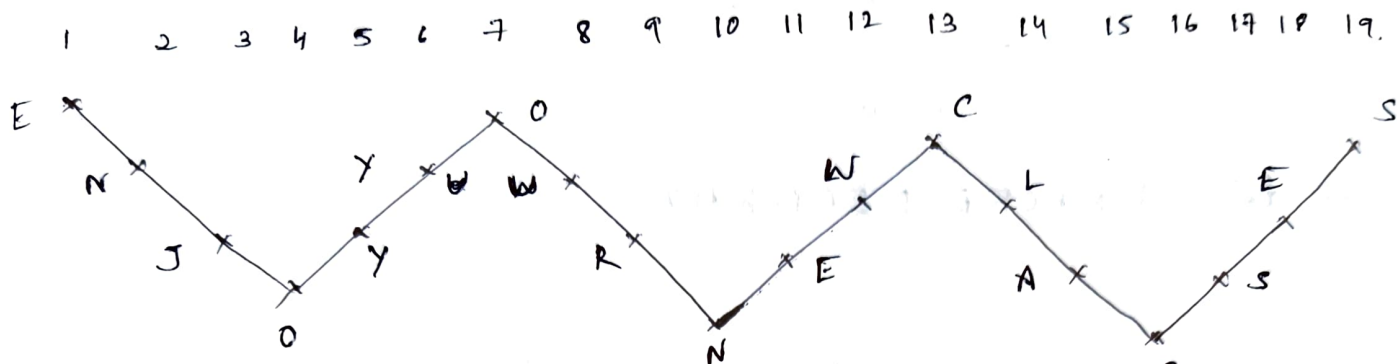
Cipher: G O D Y O D A

Depth 3

Cipher: G O D O D A O Y

3. Decrypt the cipher "EOCSNYUWLEJYREASONS", encrypted using Rail fence with depth 4

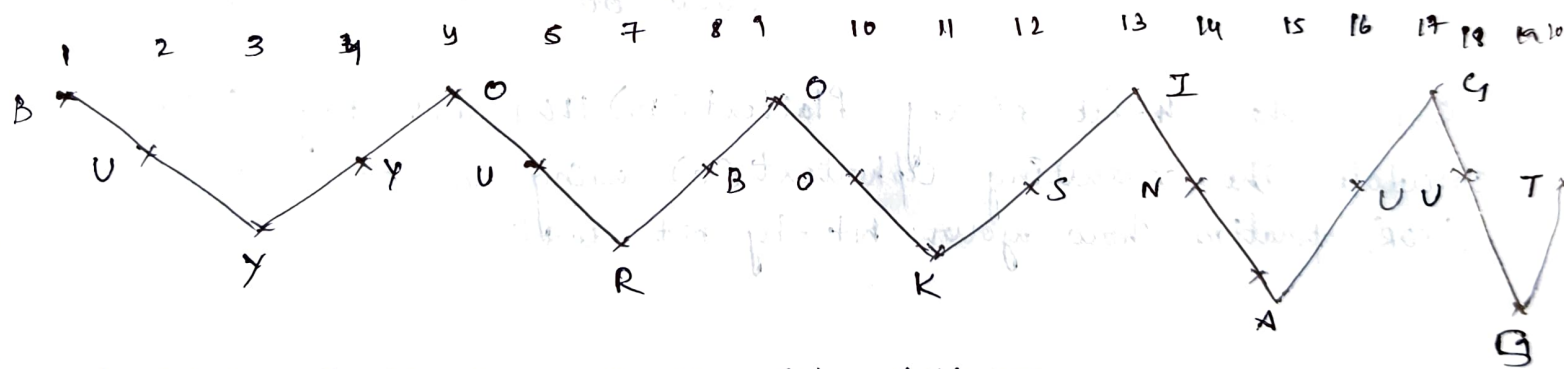
Soln: Depth 4 length: 19



plaintext: ENJOY YOUR NEW CLASSES

4. Decrypt the cipher "BOOI4UYUBOSNUUTYRKAS", encrypted using Rail fence with depth 3

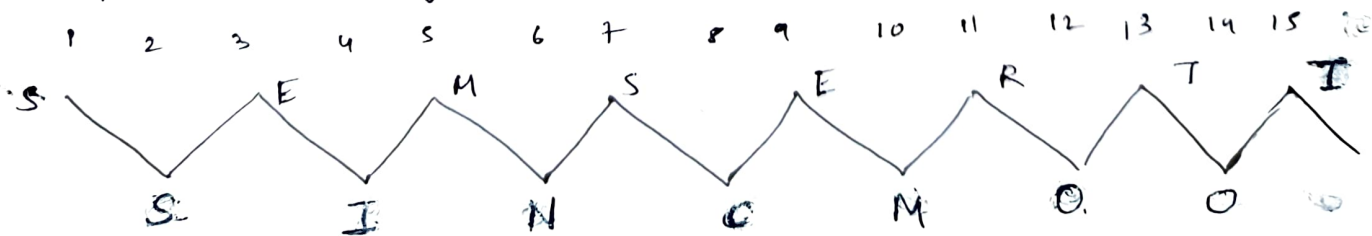
Soln: Depth: 3 length: 20



plaintext: BUY YOUR BOOKS IN AUGUST

5. Decrypt the cipher "SEMSERTISINCMOO", encrypted using Rail fence with depth 2.

Soln: Depth 2 length: 16



plaintext: SSEIMNSCEMROTOI

Permutation

1. Encrypt the following message "We are discovered" using the Key "53124"

S 3 1 2 4
W E A R E
D I S C O
V E R E D

cipher text : ASRRCEEIEEODWDV

DES

1. Using the initial permutation table, determine the output of the initial permutation box when the input is given in hexadecimal as: 0x0002 0000 0000 0001

So, hexadecimal : 0x0002 0000 0000 0001

Binary:

0000	0000	0000	0010
0000	0000	0000	0000
0000	0000	0000	0000
0000	0000	0000	0001

 } Input to box (plain-text)

↓ 15th

0000 0000 0000 0010 0000 0000 00000000 0000 0000 0000 000000000000

0000

Input bit 64 = 1, output position 25 = 1

Input bit 15 = 1, output position 63 = 1

2 By seeing permutation table

64-bit output:

[illegible]

output² in hexadecimal :

0x0000000800000002

2.) Given the 4-bit Plaintext (P) 1101 and Key (K) 1011, calculate the resulting ciphertext (C) using a bitwise Exclusive-OR (XOR) operation. Show your bit-by-bit work.

Sol: Plaintext : 1101
Key : 1011

XOR		
A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

$$\begin{array}{r} 1101 \\ 1011 \\ \hline 0110 \end{array}$$

cipher text : 0110

3. The input to S-box 1 is 000000. What is the output

Sol. 000000
 $\begin{array}{c} \uparrow \quad \uparrow \\ 1^{st} \quad 6^{th} \text{ bit} \end{array}$
 $\left\{ \begin{array}{l} 1^{st} \text{ \& } 6^{th} \text{ bit} \rightarrow \text{row} \\ \text{Middle 4 bits} \rightarrow \text{Column} \end{array} \right.$

So, 0th Row & 0th column see the number in S-box $\rightarrow 14 //$

Binary conversion of 14 $\rightarrow 1110$

output = 1110